Fizika

9 sinf

MAVZU: Masalalar yechish

Ideal gaz

$$\overline{\vartheta^2} = \frac{\vartheta_1^2 + \vartheta_2^2 + \cdots + \vartheta_N^2}{N}$$

$$p = \frac{1}{3} \cdot n \cdot m_0 \cdot \overline{\vartheta^2} = \frac{1}{3} \cdot \rho \cdot \overline{\vartheta^2}$$

$$p = \frac{2}{3} \cdot n \cdot \overline{E_k} \; dan$$

$$\overline{E_k} = \frac{3 \cdot p}{2 \cdot n}$$

Temperatura – moddaning issiqlik darajasini miqdor jihatidan aniqlaydigan fizik kattalik.

$$T = t + 273$$

Temperatura – gaz molekulalari ilgarilanma harakatining o‘rtacha kinetik energiyasining o‘lchovidir.

$$\overline{E_k} = \frac{3}{2} \cdot k \cdot T$$

$$p = \frac{2}{3} \cdot n \cdot \overline{E_k} \qquad p = n \cdot k \cdot T$$

$$\overline{\vartheta} = \sqrt{\frac{3 \cdot R \cdot T}{M}}$$

2-namunaviy masala

Qanday temperaturadagi vodorod molekulalarining o‘rtacha kvadratik tezligi 580 K temperaturadagi geliy gazi molekulalarining o‘rtacha kvadratik tezligiga teng bo‘ladi?

Berilgan:

$$M_1 = 2\,\frac{g}{mol} = 0{,}002\,\frac{kg}{mol}$$

$$M_2 = 4\,\frac{g}{mol} = 0{,}004\,\frac{kg}{mol}$$

$$T_2 = 580\ \text{K}$$

$$\overline{\vartheta_1} = \overline{\vartheta_2}$$

Formula:

$$\overline{\vartheta^2} = \frac{3 \cdot R \cdot T}{M}$$

$$\overline{\vartheta_1} = \overline{\vartheta_2} \Rightarrow \overline{\vartheta_1^2} = \overline{\vartheta_2^2}$$

$$\frac{3 \cdot R \cdot T_1}{M_1} = \frac{3 \cdot R \cdot T_2}{M_2}$$

$$T_1 = \frac{M_1 \cdot T_2}{M_2}$$

Topish kerak: $T_1 - ?$

Yechish:

$$T_1 = \frac{0{,}002\frac{kg}{mol} \cdot 580\ \text{K}}{0{,}004\ kg/mol} = 290\ \text{K}$$

Javob: $T_1 = 290\ \text{K}$

6-mashq 3-masala

Gaz molekulalari ilgarilanma haraklat tezligining o‘rtacha kinetik energiyasi $9{,}52 \cdot 10^{-21}$ J bo‘lgan gazning temperaturasini aniqlang.

Berilgan:

$E_k = 9{,}52 \cdot 10^{-21}\text{J}$

$k = 1{,}38 \cdot 10^{-23}$ J/K

Formula:

$E_k = \frac{3}{2} \cdot k \cdot T$

$T = \frac{2 \cdot E_k}{3 \cdot k}$

Topish kerak: $T - ?$

Yechish: $\text{T} = \frac{2 \cdot 9{,}52 \cdot 10^{-21}\ J}{3 \cdot 1{,}38 \cdot 10^{-23}\ J/K} \approx 460\ K$

Javob: $\mathbf{T = 460\ K}$

6-mashq 5-masala

Bir atomli gaz molekulalarining ilgarilanma harakat o‘rtacha kinetik energiyasi $1{,}2 \cdot 10^{-20}\,\mathrm{J}$ va bosimi 2,4 MPa bo‘lsa, shu gaz molekulalarining konsentratsiyasi qanday bo‘ladi?

Berilgan:

$\overline{E_k} = 1{,}2 \cdot 10^{-20}\text{J}$

$p = 2{,}4\ \text{MPa} = 2{,}4 \cdot 10^{6}\text{Pa}$

Formula:

$p = \frac{2}{3} \cdot n \cdot \overline{E_k}$

$n = \frac{3 \cdot p}{2 \cdot \overline{E_k}}$

Topish kerak: $n - ?$

Yechish: $n = \frac{3 \cdot 2{,}4 \cdot 10^{6} Pa}{2 \cdot 1{,}2 \cdot 10^{-20} J} = 3 \cdot 10^{26} m^{-3}$

Birligi: $Pa = \frac{N}{m^2}; J = N \cdot m; \frac{Pa}{J} = \frac{N}{m^2 \cdot N \cdot m} = \frac{1}{m^3} = m^{-3}$

Javob: $n = 3 \cdot 10^{26} m^{-3}$

6-mashq 6-masala

Miqdori 2 mol bo‘lgan gazning idish devorlariga beradigan bosimi 10 kPa ga teng. Gaz egallab turgan hajmni aniqlang. Gazning temperaturasi 300 K.

Berilgan:

$\nu = 2\ mol$

$p = 10\ kPa = 10000\ Pa$

$T = 300\ K$

$R = 8{,}31\ \frac{J}{K \cdot mol}$

Topish kerak: $V - ?$

Formula:

$p = n \cdot k \cdot T$

$n = \frac{N}{V};\quad N = \nu \cdot N_A;$

$n = \frac{\nu \cdot N_A}{V}$

$p = \frac{\nu \cdot N_A}{V} \cdot k \cdot T\ ;\ N_A \cdot k = R$

$p = \frac{\nu \cdot R \cdot T}{V} \Rightarrow V = \frac{\nu \cdot R \cdot T}{p}$

Yechish:

$$V = \frac{2\ mol \cdot 8{,}31\ \frac{J}{K \cdot mol} \cdot 300\ K}{10000\ Pa} \approx 0{,}5\ m^3$$

$$\frac{Pa}{J} = \frac{1}{m^3} \Rightarrow \frac{J}{Pa} = m^3$$

Javob: $V = 0,5\ m^3$

3-namunaviy masala

Gaz temperaturasi 150 K ga oshirilganda, molekulalarining o‘rtacha kvadratik tezligi 250 m/s dan 500 m/s gacha ortdi. Gazning dastlabki temperaturasi qanday bo‘lgan?

Berilgan:

$T_2 = T_1 + \Delta T$

$\Delta T = 150$ K

$\upsilon_1 = 250$ m/s

$\upsilon_2 = 500$ m/s.

Topish kerak:

$T_1 = ?$

Formulasi:

$$\upsilon_1 = \sqrt{\frac{3 \cdot R \cdot T_1}{M}};$$

$$\upsilon_2 = \sqrt{\frac{3 \cdot R \cdot T_2}{M}} = \sqrt{\frac{3 \cdot R \cdot (T_1 + \Delta T)}{M}};$$

$$\frac{\upsilon_2}{\upsilon_1} = \sqrt{\frac{T_1 + \Delta T}{T_1}}; \qquad \left(\frac{\upsilon_2}{\upsilon_1}\right)^2 = \frac{T_1 + \Delta T}{T_1};$$

$$T_1 = \frac{\Delta T}{\left(\frac{\upsilon_2}{\upsilon_1}\right)^2 - 1}. \qquad [T_1] = \frac{\text{K}}{\left(\frac{\text{m/s}}{\text{m/s}}\right)^2} = \text{K}.$$

Hisoblash:

$$T_1 = \frac{150\text{K}}{\left(\frac{500}{250}\right)^2 - 1} = 50\text{K}.$$

Javob: $T_1 = 50$ K.

Mustaqil bajarish uchun topshiriqlar

1. 6-mashq (29-sahifa) 4-masala: Molekulalar konsentratsiyasi $4 \cdot 10^{26} m^{-3}$ va bosimi $1{,}6 \cdot 10^{5} Pa$ ga teng bo‘lgan gaz molekulalarining o‘rtacha kinetik energiyasini toping.

2. 6-mashq (29-sahifa) 7-masala: Qanday temperaturadagi geliy molekulalarining o‘rtacha tezligi 350 K temperaturadagi vodorod molekulalarinig o‘rtacha kvadratik tezligiga teng bo‘ladi?

3. 6-mashq (29-sahifa) 8-masala: Gaz temperaturasi 150℃ ga ortganda gaz molekulalarining o‘rtacha kvadratik tezligi 300 m/s dan 600 m/s gacha ortdi. Gazning dastlabki temperaturasi qanday bo‘lgan?